

Aarushi Basu

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MSc student in Cognitive & Computational Neuroscience with research experience spanning neural circuit modelling, EEG/LFP signal processing, and neuroimaging-based disease classification. Working to develop computational models approximating disrupted brain function in psychiatric conditions, grounded in both computational theory and clinical data.

RESEARCH INTERESTS

Predictive coding · Neural circuit modelling · Neuroimaging · Neurodegenerative disease dynamics · Biologically plausible machine learning · Computational psychiatry

EDUCATION

MSc Cognitive and Computational Neuroscience | University of Sheffield, UK Sept 2025-Sept 2026

Relevant Modules: Neural Dynamics, Systems Neuroscience, Neurocognitive Modelling, Scientific Programming, Fundamentals of Neuroscience & Cognition, Research Methods

Postgraduate Diploma in Advanced Studies and Research | Ashoka University, India June 2024 - May 2025

Capstone thesis: MRI-Based Detection of Alzheimer's disease using Convolutional Neural Networks CGPA 3.20

PEER REVIEWED PUBLICATIONS & POSTERS

1. Sarthak Harne, Monjoy Narayan Choudhury, **Aarushi Basu**, et al. 2024. CASE: Efficient Curricular Data Pre-training for Building Assistive Psychology Expert Models. Findings of the Association for Computational Linguistics: **EMNLP 2024**, Miami, Florida, USA. Association for Computational Linguistics (ACL).
2. *Poster Abstract*: Arun Poonambalam, Krishnan Venkitashwaran, **Aarushi Basu**, et al. 2025. Graph-ESNS for cortical coordination: mapping neural perspectives in a shared embedding space. Program No. PST145.05. 2025 Neuroscience Meeting Planner. San Diego, CA. **Society for Neuroscience**, 2025. Online

RESEARCH EXPERIENCE

Research Intern — Clinical Neuroscience Jul 2024 -Aug 2024

Institute of Neurosciences, Kolkata (Dr. Hrishikesh Kumar)

- Assisted Dr. Kumar in observing and documenting clinical management of movement disorders (Parkinson's disease, SCA-12, stroke, encephalopathy) across an 8-week placement; administered MoCA assessments and gait tests, and assisted with botulinum toxin administration in focal dystonia patients in a hospital setting.
- Gathered patient histories and learned analysis of neuroimaging EEG, MRI and CT scans at the clinical level.
- Built a working framework for mapping clinical symptom profiles onto underlying neural mechanisms ; experience that informs how I construct computational models around clinically observable deficits.

Research Intern — NLP & Mental Health AI

International Institute of Information Technology, Bangalore (Supervised by Prof. T.K. Srikanth & Prof. Madhav Rao.)

Jun 2023 -Aug 2023

- Contributed to CASE-BERT, a curricular pre-training framework for BERT-based mental health disorder detection, achieving F1 = 0.91 (Depression) and F1 = 0.88 (Anxiety) on the CounselChat benchmark; outperforming MentalBERT and all other baselines (discriminative and generative) while using less than 1/37th the pre-training data (7.57M vs. 2.87B words). Responsible for data pre-processing, model fine-tuning, and evaluation pipeline across CounselChat, Depression_Reddit (F1 = 0.965), and Dreddit benchmarks; *work published at EMNLP 2024*.

RESEARCH PROJECTS

MSc Dissertation: Beyond Metacognitive accuracy: a computational account of belief updating in Schizophrenia. |

Feb 2026 - Sep 2026

University of Sheffield - Supervised by Dr. Hazem Toutounji and Dr. Alekhya Mandal

- Designed a computational pipeline that fits a Drift Diffusion Model and Two Stage Dynamic Signal Detection Model to trial-by-trial confidence ratings, quantifying belief updating as a behavioural learning rate in schizophrenia.
- Developed analysis of the differential dynamics involved in confidence and decision making in healthy control and schizophrenia spectrum disorder patients.

Graph Echo State Network for Gamma-Band LFP Modelling | SfN 2025 Poster

June 2025

Society for Neuroscience Annual Conference

- Identified a limitation of standard Echo State Networks in resolving frequency-specific gamma dynamics, and addressed it by designing a G-ESN with resonator bank extensions and multi-head attention, enabling the model to separately track low-gamma (30–80 Hz) and high-gamma (80–300 Hz) sub-bands across multi-channel LFP recordings.
- Presented as a Poster Abstract at the Society for Neuroscience Annual Conference (SfN 2025, San Diego, CA).
<https://www.abstractsonline.com/pp8/#!/21171/presentation/29625>

Capstone thesis: MRI-Based Detection of Alzheimer's disease using Transfer learning in Convolutional Neural Networks. Ashoka University | Supervised by Prof. Anirban Sen and Dr. Dipanjan Ray.

Sept 2024 -Dec 2024

- Built a curriculum transfer learning pipeline classifying MRI scans across three diagnostic categories (AD, MCI, and healthy controls), achieving 95% accuracy and $F1 = 0.97$ on AD vs. normal classification. Analysed layer-wise feature activations to identify representations driving diagnostic discrimination, finding mid-level texture features in hippocampal and entorhinal cortex most predictive of MCI-to-AD progression.

Computational Neuroscience Workshop | CAMP Workshop, IISER Pune

Jul 2025

- Implemented Wilson-Cowan neural mass models to simulate theta-gamma coupling dynamics, demonstrating that theta-gamma nesting is critically dependent on the E/I ratio, with modest shifts in inhibitory gain collapsing stable coupling into desynchronised states.
- Separately modelled dendritic pruning effects on stellate cell firing, finding that progressive pruning beyond a threshold reduced intrinsic excitability in patterns consistent with synaptic loss in neurodevelopmental conditions; work presented to faculty including Dr. Rishikesh Narayanan, Dr. Suhita Nandkarni, Dr. Collins Assisi, and Dr. Abhilasha Joshi.

TECHNICAL SKILLS

- **Programming:** Python , MATLAB
- **Deep Learning Frameworks:** PyTorch, TensorFlow
- **Neuroimaging:** MRICron, CONN toolbox
- **Computational Modelling:** RNNs, Rescorla–Wagner models, Wilson–Cowan neural mass models, Echo State Networks, CNNs, NLP models, Izhikevich neural population models.
- **Data Analysis:** Statistical analysis, clustering, classification, EEG/LFP signal processing, MRI data processing, neuroimaging data pipelines · **Visualisation:** matplotlib, seaborn, pandas
- **Clinical Assessment:** MoCA administration, gait assessment, patient-facing clinical research, diagnostic analysis

CERTIFICATIONS

- **Fundamentals of Neuroscience and Neuroimaging** | Johns Hopkins University (Coursera, Jan 2025)
- **Linear Algebra, Linear Systems and Matrix Equations** | Johns Hopkins University (Coursera, Feb 2025)

- **AI for Medical Diagnosis** | DeepLearning.ai (Coursera, Nov 2024)